

EE 576 Project 3: Object Representation and Recognition

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Abstract—This project uses SIFT Bag-of-Words for bowl, coffee mug, and soda can recognition. The method is tested on object images and kitchen scene images, and YOLOv8n is used as a baseline. Results are reported with Precision-Recall curves.

I. INTRODUCTION AND METHODOLOGY

The selected classes are *bowl*, *coffee_mug*, and *soda_can*. A deterministic 75/25 split was used. The train/test counts are 435/145 for bowl, 411/137 for coffee mug, and 471/157 for soda can.

For Part 1.a, SIFT descriptors were extracted inside the object masks. A BOW vocabulary with 80 words was learned by k-means. Each image was represented by a normalized histogram, and each class model was the average training histogram [3], [4].

For Part 1.b, object test images were scored with Chi-square histogram distance, using $1/(1 + d)$.

For Part 1.c, scene candidates were generated with thresholding, morphology, and connected components, then classified with the same BOW models [1], [2].

For Part 1.d, YOLOv8n was used with OpenCV DNN. The COCO mappings were bowl=45, coffee mug→cup=41, and soda can→bottle=39. For scene images, detections were matched with ground truth using $\text{IoU} \geq 0.30$.

The assignment asks for three τ values. Here, nine values were used: $\tau = 0.10, 0.20, \dots, 0.90$. This gives a denser Precision-Recall curve while still including three representative levels. In the plots, $\tau = 0.10$, $\tau = 0.50$, and $\tau = 0.90$ are marked with square markers. The best point is the closest one to (1, 1).

II. RESULTS

Table I gives the best point of each curve. Figures 1–5 show the required Precision-Recall plots. Square markers show the representative τ values.

III. EVALUATION AND DISCUSSION

BOW object results: BOW works well on object images. Bowl is the best class with distance 0.068, and soda can is also good with 0.175. Coffee mug has high recall but low precision, so it gives many false positives.

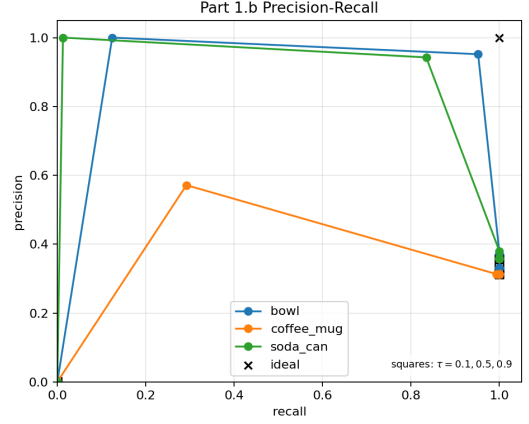


Fig. 1: Part 1.b Precision-Recall curve for BOW object recognition.

BOW scene results: Scene BOW results are weak because detection depends on automatic candidate boxes. The thresholding and connected-component step often produces boxes with background, merged objects, or partial objects. These boxes are different from the clean masked object images used in training, so many BOW scores become unreliable.

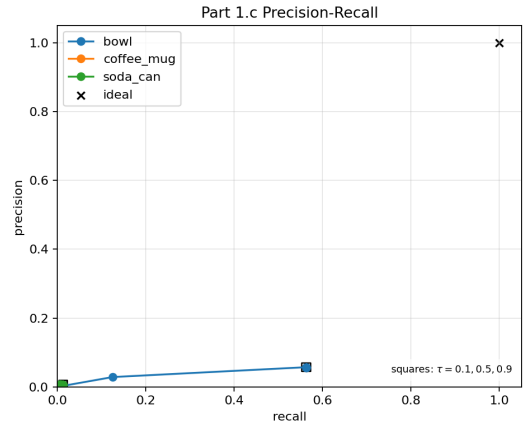


Fig. 2: Part 1.c Precision-Recall curve for BOW recognition on automatically segmented scene candidates.

Diagnostic scene result: A diagnostic test was added to check this problem. In this test, ground-truth boxes were used only as candidate regions, but recognition still used the same BOW models. This is not the main Part 1.c result. It only

TABLE I: Best Precision-Recall operating points. Dist. is the distance to the ideal point (1, 1).

Part	Class	τ	TP	FP	FN	Precision/Recall	Dist.
1.b BOW object	bowl	0.70	138	7	7	0.952 / 0.952	0.068
1.b BOW object	coffee mug	0.60	136	298	1	0.313 / 0.993	0.687
1.b BOW object	soda can	0.70	131	8	26	0.942 / 0.834	0.175
1.c BOW scene	bowl	0.10	54	890	42	0.057 / 0.562	1.039
1.c BOW scene	coffee mug	0.60	1	263	88	0.004 / 0.011	1.404
1.c BOW scene	soda can	0.10	2	346	151	0.006 / 0.013	1.401
1.d YOLO object	bowl	0.10	79	0	66	1.000 / 0.545	0.455
1.d YOLO object	coffee mug	0.40	112	1	25	0.991 / 0.818	0.183
1.d YOLO object	soda can	0.10	86	0	71	1.000 / 0.548	0.452
1.d YOLO scene	bowl	0.30	28	16	68	0.636 / 0.292	0.796
1.d YOLO scene	coffee mug	0.50	72	41	17	0.637 / 0.809	0.410
1.d YOLO scene	soda can	0.30	120	211	33	0.363 / 0.784	0.673

separates segmentation error from recognition error. For bowl, precision/recall improved from 0.057/0.562 to 0.447/0.875, so candidate generation is a major reason for the poor Part 1.c curve.

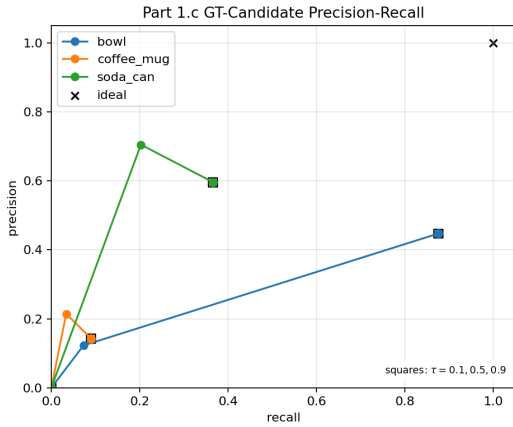


Fig. 3: Diagnostic Precision-Recall curve using ground-truth scene boxes as candidates.

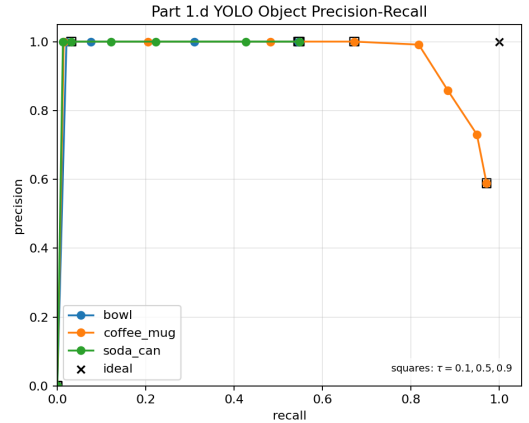


Fig. 4: Part 1.d Precision-Recall curve for YOLO on object images.

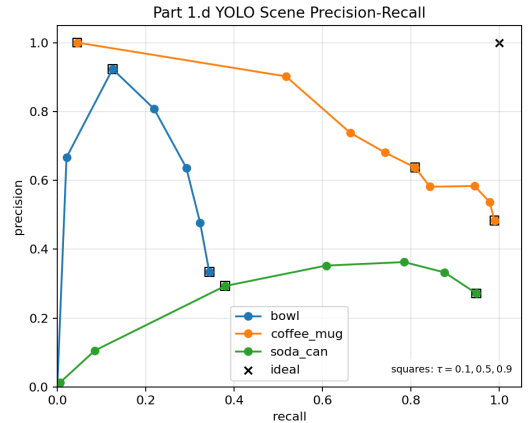


Fig. 5: Part 1.d Precision-Recall curve for YOLO on kitchen scene images.

YOLO results: YOLO gives better scene results than the classical pipeline. Coffee mug is the best YOLO scene class with distance 0.410. Soda can is harder because it is evaluated through the COCO bottle class.

Class comparison: Bowl is the best class for BOW object recognition. Coffee mug is the best class for YOLO scene detection.

IV. CONCLUSION

BOW gives useful results on isolated object images, especially for bowl and soda can. For scenes, the classical method is limited by segmentation quality. YOLOv8n gives stronger scene results because it directly predicts semantic boxes.

AI USAGE DECLARATION

ChatGPT/Codex (April 2026) was used as a supportive tool for code implementation and debugging. The final code was reviewed and tested line by line, and the reported results were verified.

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